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Feature-level SMOTE: Augmenting Fault Samples in the Learnable Feature Space for Imbalanced Fault Diagnosis of Gas Turbines

Abstract

A challenge in fault diagnosis of gas turbines is that monitoring data is composed of a small number of fault samples and an overwhelmingly large number of normal samples. Conventional data augmentation techniques generate fault samples in original data spaces, resulting in the issue that synthetic fault samples highly overlap with normal samples. To address the issue, this paper develops a novel feature-level data augmentation method, namely feature-level SMOTE, which integrates the deep Siamese multi-head self-attention network (DSMHSA) with synthetic minority over-sampling technique (SMOTE) to reduce class imbalance and overlap simultaneously. First, the DSMHSA maps the original data into a feature space, in which the inter-class samples are far away from each other while the intra-class samples are close to each other. Second, the SMOTE generates synthetic fault samples in the learned feature space, in order to balance the data set. Finally, the performance of the developed feature-level SMOTE in imbalanced fault diagnosis has been evaluated using actual monitoring data of gas turbines.

Keywords:
gas turbines, imbalanced data, inter-class overlap, feature-level data augmentation, fault diagnosis

1. Introduction

Gas turbines are the core subsystem of airplanes and work under extremely harsh conditions with high temperature, high pressure, and high speed. Their components will inevitably encounter performance degradation or even failure, such as exhaust gas temperature indication faults (EIF) and total air temperature sensor indication faults (TSIF), which will increase the probability of flight accidents [1][2]. Early fault diagnosis of gas turbines is an important way to avoid fatal breakdowns [3]. That is, if faults can be diagnosed in time, the safe operation of gas turbines can be ensured through maintenance [4]. Therefore, fault diagnosis for gas turbines has become one of the important measures in gas turbine health management.

Fault diagnosis methods for gas turbines can be mainly divided into three categories: model-based methods [5], [6], conventional machine learning-based methods [7], [8], and deep learning-based methods. First, most model-based methods require an accurate physical simulation model to estimate the performance of gas turbines under specific operating conditions. For example, Simon [5] calculates residuals between the measured values of the gas turbine and the estimated values of the physical simulation model, and thus infers whether the gas turbine fails or not. However, the damage propagation process and dynamic response of gas turbines are very complex, which would make the whole modeling process very arduous [9]. Second, the conventional machine learning methods require experienced engineers to manually select the statistical features of the monitoring data. For example, Ma et al [7] select eleven statistical features of the monitoring data in the time and frequency domains, and evaluate the performance degradation of gas turbines in combination with the subspace-based minimum volume ellipsoid method. However, these methods are at risk of failure if the selected features cannot accurately characterize the health status of gas turbines [10].

In contrast, deep learning-based methods with strong nonlinear mapping ability can automatically learn useful features from the monitoring data [11][12], and then use these features to infer the health status of gas turbines. The whole process requires neither a complex modeling process nor an engineer to specifically select statistical features, which makes the fault diagnosis task more straightforward [13].

However, deep learning-based methods for gas turbine fault diagnosis still face the following two dilemmas:

1) **Class imbalance:** In engineering practice, the labeled fault samples are relatively precious and much less than normal samples [14], as depicted in Fig.1(a). Notably, the optimization goal of deep learning methods is to minimize classification error. Normal samples will have a dominant position in classification error, and deep learning methods tend to favor normal samples and ignore fault samples during the training process. The result

is that these methods still yield relatively low classification errors, even when all fault samples are misclassified into the normal class. In this case, diagnosis accuracy over fault samples will be poor.

2) **Inter-class overlap:** When gas turbines suffer incipient faults, fault samples are close to recent normal samples, i.e., these fault samples highly overlap with normal samples, as depicted in Fig.1(a). The probability density of normal samples is higher than that of fault samples in the overlapping area. Deep learning methods tend to diagnose fault samples as normal samples, which also leads to poor diagnosis results for fault samples [15]. Therefore, to further enhance the diagnosis performance of fault samples, two priorities are to achieve the balance between different classes and reduce the degree of inter-class overlap.

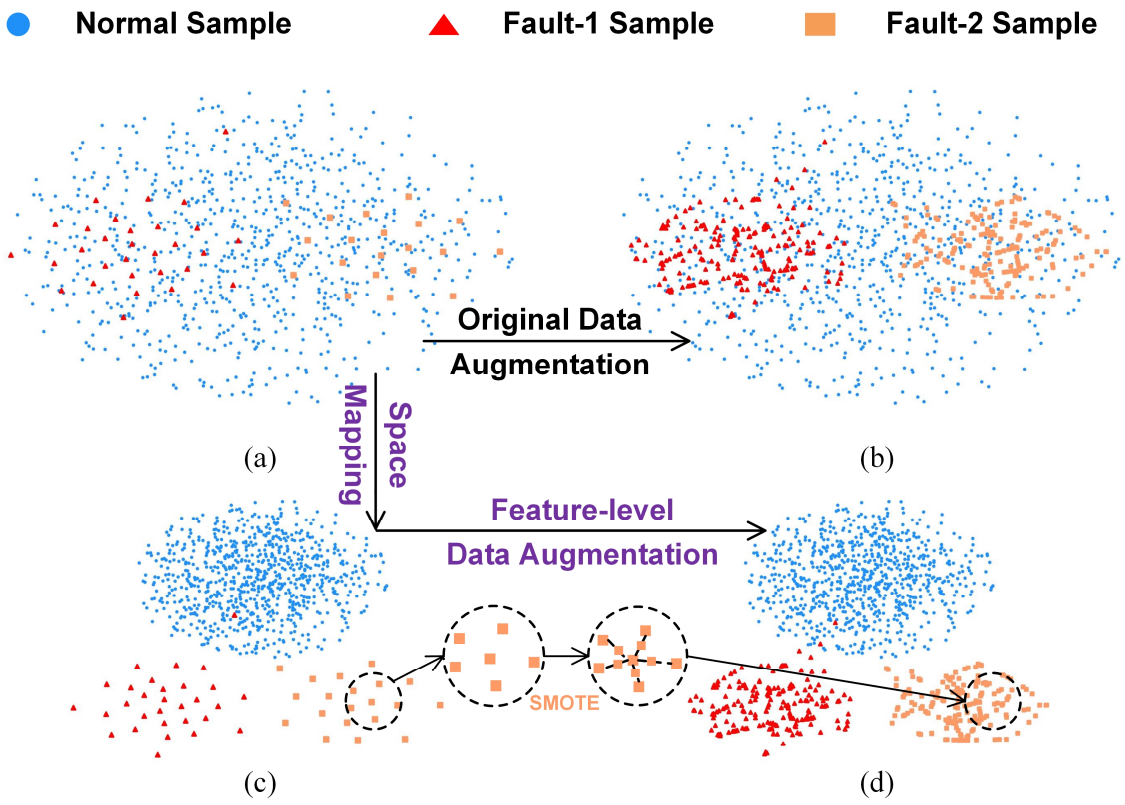

Fig. 1 2D sketch maps of (a) samples in a traditional feature space, (b) samples after data augmentation in a traditional feature space, (c) samples in the learned feature space using the DSMHSA, (d) samples after data augmentation in the learned feature space.

Up to now, data augmentation techniques are still frequently used methods to balance the dataset, especially SMOTE and its variants [16]-[25]. The classical SMOTE generates synthetic samples by randomly sampling on the lines between minority-class samples and their nearest neighbors [16]. Then, the sample distribution becomes like that in Fig.1(b), which is obviously more desirable than that in Fig.1(a). It can be observed that the probability density of fault samples in the central part of the overlap area is higher than that of normal samples, and consequently these fault samples will have higher probabilities to be correctly diagnosed.

However, the aforementioned data augmentation techniques still suffer from the following weaknesses. On one hand, it can be seen from Fig.1(b) that the probability density of normal samples in the central part of the overlap area is lower than that of fault samples, thus these normal samples face the risk of being diagnosed as fault samples. That is, the accuracy improvement of fault samples is at the expense of some normal samples. On the other hand, the probability density of fault samples in the border part of the overlap area is still lower than that of normal samples, and these fault samples tend to be misdiagnosed as normal samples [26].

Aiming at the weaknesses of traditional data augmentation methods, this paper develops a novel feature-level data augmentation method. The major contributions of this paper can be summarized as below.

1) A feature space with large inter-class distances is learned via deep Siamese multi-head self-attention network (DSMHSA). The DSMHSA can map the original data set into a feature space as depicted in Fig.1(c), in which samples from the same class are close to one another, while samples from different classes are far away from one another [27]. This is achieved by minimizing intra-class distances and maximizing inter-class distances. Besides, the multi-head self-attention network (MHSANet) is selected as the feature extraction module of the Siamese architecture, which can effectively deal with multi-dimensional time series.
2) Synthetic fault samples are generated in the learned feature space via synthetic minority over-sampling technique (SMOTE). Synthetic fault samples are put together with original samples to serve as the balanced dataset, and the sample distribution may become like that indicated in Fig.1(d), which is obviously more desirable than that in Fig.1(c). The reason is that fault samples will have a relatively equal position with normal samples in the

classification error, and deep learning methods tend to perform equally on samples in different class. In such a case, fault samples will be diagnosed much more easily.

3) The effectiveness of the proposed framework is verified using real-life operation data rather than simulated data. The data is collected from gas turbines of an Asian airline, in which one normal condition and two typical fault conditions (i.e., EIF and TSIF) are contained.

The rest of this paper is organized as follows. Section 2 introduces the related research works. Section 3 describes the developed feature-level SMOTE and its application in imbalanced fault diagnosis of gas turbines. The experimental results are given in Section 4. Finally, Section 5 gives the main conclusions and provides an outlook for future research work.

2. Related Work

2.1. Deep learning for gas turbine fault diagnosis

In recent years, some scholars have focused on deep learning for gas turbine fault diagnosis [28]-[31]. For example, Yuan et al. [28] applied the long short-term memory network to learn the temporal information from historical monitoring data, thus yielding high fault diagnosis and performance prediction accuracy for gas turbines. Li et al. [29] adopted a back-propagation neural network to extract performance deviation features and then realized gas turbine fault diagnosis under small sample size conditions. Khorasani et al. [30] proposed a hybrid fault diagnosis framework using the recurrent neural network and self-organizing map, which can effectively deal with coupled faults in gas turbines. Zhou et al. [31] adopted the spatial transformer network to convert different boundary conditions into designed standard boundary conditions, and then combined it with the classifier to accomplish gas turbine fault diagnosis under complex boundary conditions. Researches mentioned above require building a class-balanced training set for satisfactory results, however, the actual situation is not the case because of the lack of fault samples [32]. To address the issue, we develop a novel feature-level data augmentation method and then put it together with the classification model to improve the fault diagnosis accuracy of gas turbines.

2.2. Traditional re-sampling methods for imbalanced classification

Resampling methods are still common ways to deal with the imbalanced classification problem, which can be mainly divided into two different categories: under-sampling and over-sampling techniques [33]. Under-sampling techniques reduce the number of samples in majority class to match that of samples in minority class, and this strategy is prone to lead to information loss [33]. Aiming at the issue, many improved under-sampling techniques selectively rather than randomly remove majority class samples, including clustering-based under-sampling [34], Tomek-link based under-sampling [35], Neighborhood-based under-sampling [36]. Although these techniques try to retain the representative samples, they still cannot thoroughly avoid information loss.

In contrast, over-sampling techniques increase the number of samples in minority class to approximate that of samples in majority class, which can effectively avoid information loss [33]. The popular SMOTE generates synthetic samples of the minority class rather than random replication, which can alleviate the over-fitting problem to some extent [16]. Additionally, to further improve the quality of synthetic samples, many variants of SMOTE have emerged in recent years, including Borderline-SMOTE [17], SVM-SMOTE [18], KMeans-SMOTE [19] Safe-Level-Synthetic Minority Over-Sampling TEchnique (SL-SMOTE) [20], Density-Based Synthetic Minority Over-Sampling TEchnique (DBSMOTE) [21], Majority Weighted Minority Oversampling Technique (MWMOTE) [22], Over-Sampling Using Propensity Scores (OUPS) [23], Adaptive Semi-Unsupervised Weighted Oversampling (A-SUWO) [24], and Noise Reduction A Priori Synthetic Over-Sampling Technique (NRAS) [25]. However, the SMOTE and its variants ignore the problem that minority class overlap with majority class, and directly generating minority class samples will increase the degree of overlap. To avoid this situation, a significant task is to separate minority class from majority class before data augmentation.

3. Proposed Fault Diagnosis Framework

This section firstly introduces the selected key characteristic parameters for the gas turbine fault diagnosis. Then, the key procedures of the developed feature-level SMOTE are described in detail. Finally, the whole process of the gas turbine fault diagnosis framework is summarized.

3.1. Research motivations

Gas turbine anomaly detection is an extremely important way to ensure customer safety and save the airlines operating costs. However, deep learning-based methods face at least the following three challenges.

Data imbalance: For gas turbines, the abnormal data is very limited and far less than normal data. In the training stage, the classification loss is dominated by normal data. That is, if normal data is classified correctly, the classification loss will be very small. In this case, the classifier pays more attention to normal data [37], and tends to classify all data as normal data. However, this is not allowed in engineering.

Inter-class overlap: The abnormal data is similar to normal data (right before anomaly) and overlaps with it. In this case, the synthetic abnormal data may be closer to normal data but still has the label “abnormal”, which will mislead the training of deep learning models. Therefore, a significant task is to reduce the degree of inter-class overlap

before data augmentation.

Complex temporal pattern: The monitoring data of gas turbines belong to the typical multi-dimensional time series. The dependencies, trends, and correlations among different performance parameters are complex and unclear. That is, it is a challenging problem to extract complex temporal patterns from monitoring data.

3.2. General procedure of the proposed fault diagnosis framework

Aiming at the imbalanced issue in the case of inter-class overlap, this paper proposes a novel fault diagnosis framework that integrates the feature-level SMOTE with the classical classification model. The overall procedure of the framework is depicted in Fig.2, and the training process is mainly divided into three steps:

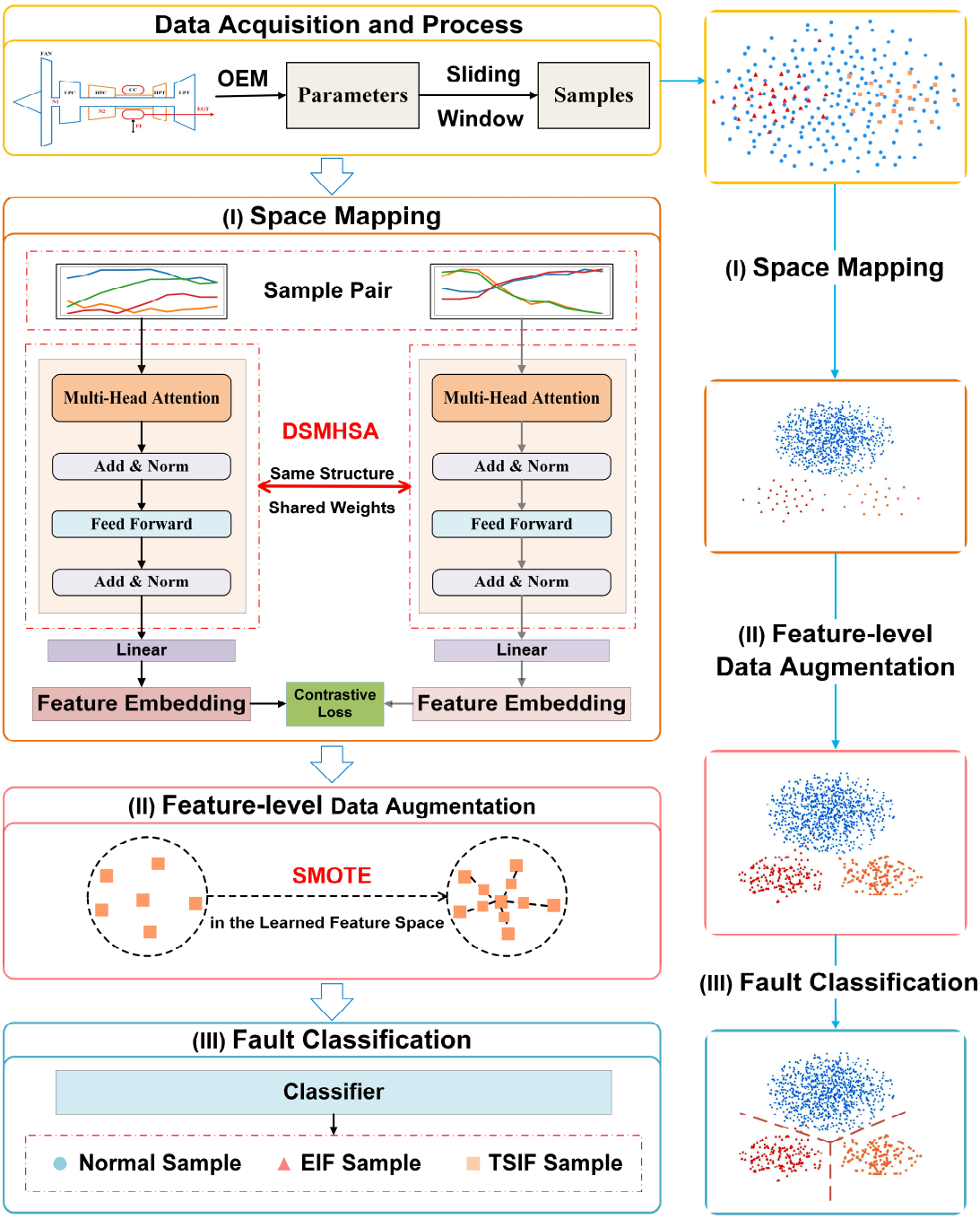

Fig 2 The procedures of the developed fault diagnosis framework.

(I) **Space Mapping:** The first step is to reduce the degree of overlap, which is realized by the DSMHSA. The DSMHSA is optimized by minimizing the contrastive loss, and the optimized DSMHSA maps samples into the feature space, in which samples of different classes are far away from one another.

(II) **Feature-level data augmentation:** The second step is to obtain a balanced training dataset, which is achieved by the SMOTE. In the learned feature space, the SMOTE is used to generate fault samples, and synthetic samples are put together with original samples to serve as the balanced training dataset.

(III) **Fault data classification:** The third step is used for fault classification, which is realized by the classification model. In the learned feature space, the classification model is trained on the balanced training dataset. Its parameters are optimized by minimizing the objective function, and then the trained classification model can be used to diagnose gas turbines.

3.3. The selected key characteristic parameters

The state characteristic parameters of gas turbines can reflect their performance status, and the extraction of fault symptoms from these parameters is an essential step for fault diagnosis. Among the numerous state characteristic parameters, the most important and effective characteristic parameters include exhaust gas temperature (EGT), low-pressure rotor speed (N1), high-pressure rotor speed (N2), and fuel flow (FF), which are selected as the key characteristic parameters for gas turbine fault diagnosis in the practical engineering.

In view of the fact that their measured values are affected by external environmental factors such as temperature and atmospheric pressure, thus measured values are usually converted to conversion values at standard atmospheric pressure of sea level ($T_0 = 288.15K, P_0 = 101.325kPa$). However, the conversion values are still affected by the operating conditions of gas turbines. It is noteworthy that baseline values represent standard values of characteristic parameters at the specific operating condition. Thus, their deviation values (i.e., differences between the conversion values and baseline values) take the specific operating condition into account, and can more accurately reflect the performance status of gas turbines.

Based on the above analysis, this paper adopts exhaust gas temperature margin (EGTM), deviation of exhaust gas temperature (DEGT), deviation of core rotational speed (DN2), and deviation of fuel flow (DFF) for gas turbine fault diagnosis, as depicted in Fig.3. Moreover, sliding window method is used to construct samples, and the construction process is as follows.

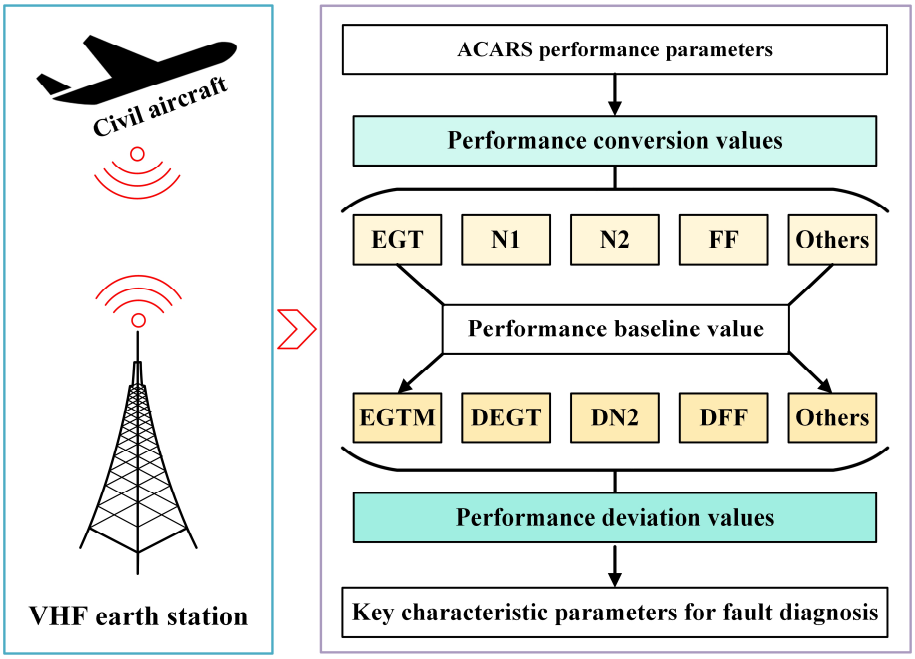

Fig 3 Acquisition and preprocessing flow of gas turbine performance data, where VHF denotes very high frequency from 30MHz to 300MHz.

First, according to maintenance reports, key characteristic parameters of the k-th gas turbine before its failure can be obtained, which are expressed by

$$KCP_k = \begin{bmatrix} EGTM_{k,1} & EGTM_{k,2} & \cdots & EGTM_{k,m_k} \\ DEGT_{k,1} & DEGT_{k,2} & \cdots & DEGT_{k,m_k} \\ DN2_{k,1} & DN2_{k,2} & \cdots & DN2_{k,m_k} \\ DFF_{k,1} & DFF_{k,2} & \cdots & DFF_{k,m_k} \end{bmatrix} \tag{1}$$

where m_k represents the total number of flight cycles of the k-th gas turbine before failure.

Then, in order to construct samples, the sliding window moves from end (i.e., the flight cycle in which fault occurs) to front. Here, the setting window and step length are set as 10. Therefore, the p-th sample of k-th gas turbine is expressed by

$$FSS_{k,p} = \begin{bmatrix} EGTM_{k,m_k-10\times p+1} & EGTM_{k,m_k-10\times p+2} & \cdots & EGTM_{k,m_k-10\times p+10} \\ DEGT_{k,m_k-10\times p+1} & DEGT_{k,m_k-10\times p+2} & \cdots & DEGT_{k,m_k-10\times p+10} \\ DN2_{k,m_k-10\times p+1} & DN2_{k,m_k-10\times p+2} & \cdots & DN2_{k,m_k-10\times p+10} \\ DFF_{k,m_k-10\times p+1} & DFF_{k,m_k-10\times p+2} & \cdots & DFF_{k,m_k-10\times p+10} \end{bmatrix} \tag{2}$$

when $p = 1$, $FSS_{k,p}$ represents the fault sample; otherwise, $FSS_{k,p}$ represents the normal sample.

Finally, to avoid the randomness of one experiment, the five-fold cross-validation is selected to divide samples. It is noteworthy that the set of normal samples (D_{NOR}), the set of EIF samples (D_{EIF}), and the set of TSIF samples (D_{TSIF}) are divided into five disjoint subsets, separately. Each time four subsets are put together as training set $D_{NOR}^{(TRAIN)}/D_{EIF}^{(TRAIN)}/D_{TSIF}^{(TRAIN)}$, and the remaining one is the test set $D_{NOR}^{(TEST)}/D_{EIF}^{(TEST)}/D_{TSIF}^{(TEST)}$. Then the training set is expressed by

$$D^{(TRAIN)} = D_{NOR}^{(TRAIN)} \cup D_{EIF}^{(TRAIN)} \cup D_{TSIF}^{(TRAIN)} \tag{3}$$

And the test set is expressed by

$$D^{(TEST)} = D_{NOR}^{(TEST)} \cup D_{EIF}^{(TEST)} \cup D_{TSIF}^{(TEST)} \tag{4}$$

3.4. Developed Feature-level SMOTE for intelligently generating synthetic fault samples

The feature-level SMOTE consists of two major steps: space mapping and minority data augmentation. The basic theory of the DSMHSA is introduced in the first place, and then procedures of minority data augmentation are described in detail.

3.4.1. Space mapping via DSMHSA

The first step of the feature-level SMOTE is to learn a space mapping from original data before data augmentation, and the space mapping can map the original data set into a feature space, in which different classes are far away from one another. In this paper, the DSMHSA is selected to learn the aforementioned space mapping.

The Siamese architecture consists of two important parts, two sub-networks with the same structure and shared weights as the feature extraction module, and the contrastive loss as the cost module. Monitoring data of gas turbines is a multidimensional time series. As an essential variant of the conventional deep neural network, the MHSANet can directly deal with multidimensional time series and play an essential role in multidimensional time series analysis in recent years [38]. Therefore, the MHSANet is selected as the feature extraction module of the Siamese architecture in this paper.

Feature extraction module: The MHSANet consists of multiple Transformer Encoder layers (i.e., a multi-head self-attention mechanism, two short-cuts & layer normalizations, and a fully connected feed-forward layer) and a linear layer. The multi-head self-attention mechanism is its key component, which can adaptively extract dependencies of different time scales. The model structure of the multi-head self-attention mechanism is depicted in Fig.4.

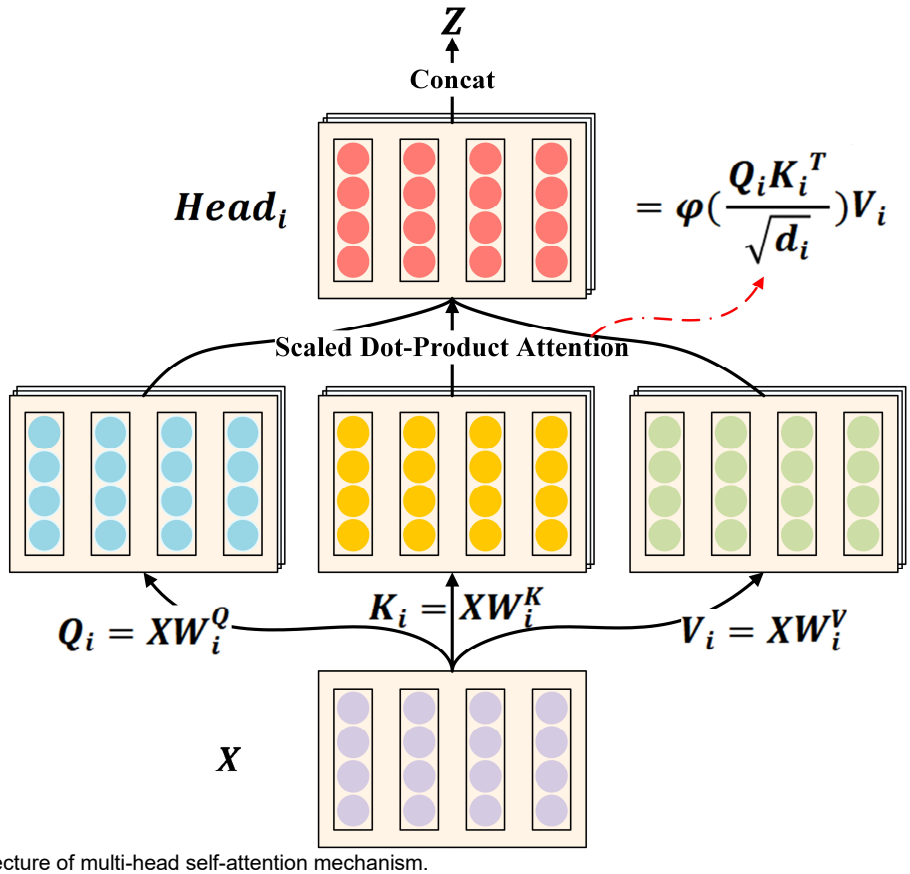

Fig 4 The architecture of multi-head self-attention mechanism.

More specifically, the calculation formulas of the multi-head self-attention mechanism are expressed by

$$Q_i = XW_i^Q, \quad K_i = XW_i^K, \quad V_i = XW_i^V \tag{5}$$

$$Att_i = \varphi(Q_i {K_i}^T / \sqrt{d_i}) \quad (6)$$

$$Head_i = Att_i V_i \quad (7)$$

$$Z = Concat(Head_1, Head_2, \cdots, Head_h) W^C \quad (8)$$

where X is the sum of input embedding and position embedding. W_i^Q, W_i^K, W_i^V represent linear projections of the i-th self-attention module, and Q_i, K_i, V_i are Query, Key and Value matrices, respectively. Then, d_i is the dimension of Q_i, and φ is the Softmax function. Att_i is the self-attention weight matrix that can enhance important features and weaken useless features. Furthermore, $Head_i$ is the output of the i-th self-attention module. Finally, W^C is the parameter matrix of the linear projection, and Z is the final feature that is obtained by the multi-head self-attention mechanism.

Cost module: The cost module adopts contrastive loss to optimize the Siamese architecture, which can effectively deal with pair data [19]. The contrastive loss is expressed by

$$L_{sia} = \frac{1}{N_{ps}} \sum_{k=1}^{N_{ps}} L_k = \frac{1}{N_{ps}} \sum_{k=1}^{N_{ps}} y_k \mathrm{D_E}(Z_k^1, Z_k^2) + (1 - y_k) max(d - \mathrm{D_E}(Z_k^1, Z_k^2), 0) \quad (9)$$

where (Z_k^1, Z_k^2) represents the k-th sample pair. Z_k^1 and Z_k^2 are the latent feature representation of X_k^1 and X_k^2, respectively. Notably, y_k is the label and its setting rule is as follows. It is set to 1 when X_k^1 and X_k^2 belong to the same class; otherwise, it is set to 0. Additionally, d is a margin and N_{ps} is the total number of sample pairs.

When minimizing the contrastive loss L_{sia}, if X_k^1 and X_k^2 belong to the same class ($y_k = 1$), the k-th term of the contrastive loss $L_k = \mathrm{D_E}(Z_k^1, Z_k^2)$ tends to be smaller and smaller, which means Z_k^1 and Z_2 are closer and closer. Conversely, if Z_k^1 and Z_k^2 belong to different classes ($y_k = 0$), then $L_k = max(d - \mathrm{D_E}(Z_k^1, Z_k^2), 0)$ tends to be smaller and smaller, i.e., $\mathrm{D_E}(Z_k^1, Z_k^2)$ is larger and larger (but not more than *d*), which means Z_k^1 and Z_k^2 are farther and farther from each other. Thus, samples from different classes tend to be far away from one another in the feature space.

3.4.2. Minority data augmentation via SMOTE

The second step of the feature-level SMOTE is to generate minority samples in the learned feature space. Up to now, the SMOTE is still one of the most efficient and practical data augmentation methods, and is widely used in the field of imbalanced classification [39]. In this paper, the SMOTE is used to generate minority samples. Then, the generation process is briefly described as follows.

First, for each minority sample z_k, its nearest latent feature representation $NN(z_k)$ is found using Euclidean distance $D_E(\cdot,\cdot)$, which is expressed by

$$NN(z_k) = \underset{z_p}{argmin}\{D_E(z_k, z_p)\}, s.t. y_k = y_p \quad (10)$$

where y_k is the label of the latent feature representation z_k, and y_p is the label of the latent feature representation z_p.

Second, the synthetic sample is generated by randomly selecting one point on the line between z_k *and* $NN(z_k)$, which is expressed by

$$GS(z_k) = \alpha z_k + (1 - \alpha) NN(z_k) \quad (11)$$

where $GS(z_k)$ is the synthetic sample and α represents a random variable that obeys the uniform distribution U[0,1].

Finally, the above steps are repeated until the generated synthetic samples can balance the minority and majority classes.

3.5. Fault data classification

After obtaining the balanced dataset in the learned feature space, appropriate classifiers are still needed for accurate fault classification. The Softmax classifier (SMC) is widely used in diagnosis and recognition problems and is chosen as the classification model [40], [41]. The classification model only consists of one input layer and one output layer. It takes the latent feature representation of samples as input and outputs the probability that samples belong to each category [40]. Then, the mean-square loss is used as the objective function to optimize the weight parameters of the classification model. Finally, the optimized classifier can be used for classifying the health status of gas turbines. It is notable that the developed feature-level SMOTE can also be combined with other classification models. For the further evaluation of the developed data augmentation method, the K-Nearest Neighbor (KNN) is also used in this paper.

4. Experimental Results

4.1. Experimental setup

4.1.1. Dataset descriptions

In this paper, monitoring data of gas turbines is collected to evaluate the effectiveness of the proposed framework. The data comes from actual operation data of an Asian airline rather than simulated data. As summarized in Table I, there are a total of 1047 samples, including 984 normal samples, 38 EIF samples, and 25 TSIF samples. The five-fold cross-validation is selected to split the training and test datasets, and each experiment is repeated five times. Unless otherwise specified, the following subsections take the first fold as the example for discussion.

TABLE I. Summary of the gas turbine dataset

Five-fold cross-validation	Number of training samples			Number of testing samples		
	Normal	EIF	TSIF	Normal	EIF	TSIF
K=1	787	30	20	197	8	5
K=2	787	30	20	197	8	5
K=3	787	30	20	197	8	5
K=4	787	31	20	197	7	5
K=5	788	31	20	196	7	5

Training the DSMHSA requires sample pairs, and the construction procedures of sample pairs are listed as follows:

First, sample pairs belonging to the same class are constructed, and the corresponding label is set as 1: (1) According to Ref. [27], each normal sample is paired with five neighboring normal samples. (2) Each EIF sample is paired with all EIF samples. The reasons are that on one hand, the number of EIF samples is relatively small; on the other hand, this can make all EIF samples as close to one another as possible. (3) Likewise, each TSIF sample is paired with all TSIF samples, and the reasons are the same.

Second, sample pairs belonging to different classes are constructed, and the corresponding label is set as 0: (1) To balance sample pairs with different labels, each EIF sample is paired with ninety neighboring normal samples. (2) For the same reasons, each TSIF sample is paired with ninety neighboring normal samples. (3) Each EIF sample is paired with all TSIF samples. The reasons are that on one hand, the numbers of EIF samples and TSIF samples are relatively small; on the other hand, this can make EIF samples as far away from TSIF samples as possible. Finally, there are 5235 sample pairs with label as 1, and 5100 sample pairs with label as 0, which are summarized in Table II.

TABLE II. Summary of the paired samples

Sample I	Sample II	Paired sample	Label	Total
Normal	Normal	<Normal, Normal>	1	787×5=3935
EIF	EIF	<EIF, EIF>	1	30×30=900
TSIF	TSIF	<TSIF, TSIF>	1	20×20=400
EIF	Normal	<EIF, Normal>	0	30×90=2700
TSIF	Normal	<TSIF, Normal>	0	20×90=1800
EIF	TSIF	<EIF, TSIF>	0	30×20=600

4.1.2. Implementation details

Experimentation is implemented using Python 3.9 with PyTorch 1.11.0 and an Intel Core i5-12400F, 16G DDR4 processor. The hyper-parameters are set according to the recommendations [42], [43]. First, the DSMHSA-related key hyperparameters include the number of Transformer Encoder layers, the number of attention heads, the dimension of the input embedding, and the dimension of the latent feature representation, which significantly affect the capability of feature extraction. The number of Transformer Encoder layers is set to 2, the number of attention heads is set to 8, the dimension of the input embedding is set to 512 and the dimension of the latent feature representation is set to 128. Second, the SMOTE-related key hyperparameter is the number of nearest neighbors, and it is set to 5. Third, the SMC has two layers. The input layer has 128 nodes, while the output layer has 3 nodes. Additionally, the number of nearest neighbors in the KNN is selected among 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 such that the metrics are averaged for the evaluation of the data augmentation methods. Finally, Adam is selected to optimize the neural network, the learning rate is 2e-4, the mini-batch size is 128, and momentums β_1 and β_2 are 0.5 and 0.999, respectively.

4.1.3. Evaluation metrics

In this paper, similar to the training set, the test set is also composed of a tiny number of fault samples and an overwhelming large number of normal samples. In this case, the classical overall accuracy has no important reference value [44]. Therefore, the confusion matrix is selected to evaluate the fault diagnosis models, which is as illustrated in Table III.

TABLE III. An illustration of confusion matrix

Actual Class	Predicted Class		
	EIF	TSIF	Normal
EIF	N_{11}	N_{12}	N_{13}
TSIF	N_{21}	N_{22}	N_{23}
Normal	N_{31}	N_{32}	N_{33}

Additionally, the individual diagnosis effect of EIF samples, TSIF samples, and normal samples can be evaluated by the recall, which are denoted as ROE, ROT, and RON, respectively. And the overall diagnosis effect of the models can be evaluated by the balanced accuracy [45], which is expressed by

$$ROE = N_{11}/(N_{11} + N_{12} + N_{13}) \tag{12}$$

$$ROT = N_{22}/(N_{21} + N_{22} + N_{23}) \tag{13}$$

$$RON = N_{33}/(N_{31} + N_{32} + N_{33}) \tag{14}$$

$$balanced\ accuracy = (RON + ROE + ROT)/3 \tag{15}$$

4.2. Validating the effectiveness of the developed method

This developed method is not only compared with a classifier without class balancing, but also compared with a number of classical data augmentation methods. The two types of comparisons are clarified as below: (1) **Baseline:** The unbalanced training set is directly used to train the classification model, and then the trained classification model is used to classify the test samples. (2) **7 data augmentation methods:** These methods, i.e., SMOTE, SL-SMOTE, DBSMOTE, MWMOTE, OUPS, A-SUWO, and NRAS, are used to balance the training dataset, respectively. Then, the classification model is trained using the balanced training dataset, and used for fault diagnosis.

The five-fold cross-validation is selected to split the training and test datasets, and thereby each cross-validation is repeated five times. The testing results of different methods combined with SMC and KNN have been summarized in Table IV and Table V, respectively, which are discussed as below.

(1) Compared with the Baseline method, it can be observed that the considered data augmentation methods combined with SMC and KNN achieve better fault diagnosis results. This illustrates the necessity of balancing the training dataset. That is to say, there are some positive benefits when using data augmentation methods.

(2) In terms of balanced accuracy, the developed method outperforms SMOTE in most cases with K=1, 2, 3, 4, 5. It indicates that the developed data augmentation strategy that generates fault samples in the learned feature space is more effective.

(3) The developed method yields the best fault diagnosis results in terms of averaged balanced accuracy. When SMC is used, compared with all comparative data augmentation methods, the averaged balanced accuracy is increased by 3.37%, 5.72%, 4.41%, 3.85%, 9.67%, 13.94%, and 12.39%, respectively. When KNN is used, the developed method achieves an average balanced accuracy improvement of 0.91%, 15.84%, 11.13%, 3.92%, 3.0%, 18.18%, and 10.98%, respectively, which further shows the usefulness of the developed method.

TABLE IV. Balanced accuracy of different fault diagnosis methods using SMC

Method	Balanced Accuracy (%)					
	K=1	K=2	K=3	K=4	K=5	Average
Baseline	33.33	33.33	33.33	33.33	33.33	33.33
SMOTE	89.05	79.23	96.16	89.94	**80.66**	87.01
SL-SMOTE	84.27	76.96	90.97	90.46	80.61	84.66
DBSMOTE	87.94	78.1	95.23	90.57	78.01	85.97
MWMOTE	88.15	77.83	95.09	92.32	79.27	86.53
OUPS	87.36	73.93	89.88	82.88	69.49	80.71
A-SUWO	73.14	79.15	81.52	79.79	68.59	76.44
NRAS	85.31	73.72	64.89	90.39	75.63	77.99
Developed	**91.07**	**87.47**	**98.22**	**95.99**	79.13	**90.38**

TABLE V. Balanced accuracy of different fault diagnosis models using KNN

Method	Balanced Accuracy (%)					
	K=1	K=2	K=3	K=4	K=5	Average
Baseline	52.23	50.9	59.91	58.55	57.33	55.79
SMOTE	**95.21**	79.68	89.1	90.64	91.85	89.3
SL-SMOTE	70.71	72.79	81.36	72.91	74.09	74.37
DBSMOTE	80.36	68.09	82.2	78.44	86.29	79.08
MWMOTE	91.96	76.91	86.06	84.75	91.77	86.29
OUPS	92.84	74.03	**94.05**	86.71	88.44	87.21
A-SUWO	65.28	73.22	74.73	79.66	67.25	72.03
NRAS	70.21	**81.87**	81.87	88.73	73.9	79.32
Developed	94.17	79.02	90.74	**94.81**	**92.33**	**90.21**

Further, the experimental results in terms of ROE, ROT, and RON are summarized in Table VI - Table XI. It can be observed that the developed method achieves superior performance in the most experimental comparison. First, according to the ROE, the developed method combined with SMC outperforms the first eight methods in the case of K=1, 2, 3, 4, and has similar performance with DBSMOTE, and MWMOTE in terms of average ROE. When the KNN is used, the developed method obtains the first three positions in the case of K=1, 2, 3, 4, 5. Second, for the ROT, the developed method combined with SMC / KNN yields at least 4.0% / 2.4% improvement compared to most of the other considered methods. It indicates that the developed method can effectively recognize fault samples.

TABLE VI. ROE of different fault diagnosis methods using SMC

Method	ROE (%)					
	K=1	K=2	K=3	K=4	K=5	Average
Baseline	0	0	0	0	0	0
SMOTE	**100**	**75**	**100**	92.86	**95.71**	92.71
SL-SMOTE	85	**75**	85	90	94.29	85.86
DBSMOTE	98.75	**75**	**100**	95.71	88.57	91.61
MWMOTE	98.75	**75**	**100**	**100**	92.86	**93.32**
OUPS	90	**75**	96.25	90	74.29	85.11
A-SUWO	47.5	**75**	50	57.14	57.14	57.36
NRAS	81.25	51.25	40	78.57	72.86	64.79
Developed	**100**	**75**	**100**	**100**	85.71	92.14

TABLE VII. ROE of different fault diagnosis models using KNN

Method	ROE (%)					
	K=1	K=2	K=3	K=4	K=5	Average
Baseline	8.75	8.75	6.25	5.71	10	7.89
SMOTE	91.25	43.75	70	77.14	81.43	72.71
SL-SMOTE	33.75	45	45	21.43	60	41.04
DBSMOTE	57.5	30	47.5	37.14	60	46.43
MWMOTE	85	42.5	60	57.14	77.14	64.36
OUPS	**98.75**	42.5	**98.75**	85.71	**82.86**	**81.71**
A-SUWO	26.25	40	25	40	38.57	33.96
NRAS	30	**47.5**	48.75	68.57	58.57	50.68
Developed	88.75	42.5	75	**91.43**	**82.86**	76.11

TABLE VIII. ROT of different fault diagnosis methods using SMC

Method	ROT (%)					
	K=1	K=2	K=3	K=4	K=5	Average
Baseline	0	0	0	0	0	0
SMOTE	80	80	**100**	**100**	**60**	84
SL-SMOTE	80	80	**100**	**100**	**60**	84
DBSMOTE	80	80	**100**	**100**	**60**	84
MWMOTE	80	80	**100**	**100**	**60**	84
OUPS	**100**	80	**100**	**100**	**60**	**88**
A-SUWO	80	80	**100**	**100**	**60**	84
NRAS	80	80	60	**100**	**60**	76
Developed	80	**100**	**100**	**100**	**60**	**88**

TABLE IX. ROT of different fault diagnosis models using KNN

Method	ROT (%)					
	K=1	K=2	K=3	K=4	K=5	Average
Baseline	48	44	74	70	62	59.6
SMOTE	96	**100**	**100**	**100**	**100**	**99.2**
SL-SMOTE	80	78	**100**	**100**	64	84.4
DBSMOTE	84	76	**100**	**100**	**100**	92
MWMOTE	92	92	**100**	**100**	**100**	96.8
OUPS	**96**	**100**	**100**	**100**	**100**	**99.2**
A-SUWO	70	82	**100**	**100**	64	83.2
NRAS	82	**100**	**100**	**100**	66	89.6
Developed	**96**	**100**	**100**	**100**	**100**	**99.2**

TABLE X. RON of different fault diagnosis methods using SMC

Method	RON (%)					
	K=1	K=2	K=3	K=4	K=5	Average
Baseline	**100**	**100**	**100**	**100**	**100**	**100**
SMOTE	87.16	82.69	88.48	76.95	86.28	84.31
SL-SMOTE	87.82	75.89	87.92	81.37	87.55	84.11
DBSMOTE	85.08	79.29	85.69	75.99	85.46	82.3
MWMOTE	85.69	78.48	85.28	76.95	84.95	82.27
OUPS	72.08	66.8	73.4	58.63	74.18	69.02
A-SUWO	91.93	82.44	94.57	82.23	88.62	87.96
NRAS	94.67	89.9	94.67	92.59	94.03	93.17
Developed	93.2	87.41	94.67	87.97	91.68	90.99

TABLE XI. RON of different fault diagnosis models using KNN

Method	RON (%)					
	K=1	K=2	K=3	K=4	K=5	Average
Baseline	**99.95**	**99.95**	**99.49**	**99.95**	**100**	**99.87**
SMOTE	98.38	95.28	97.31	94.77	94.13	95.97
SL-SMOTE	98.38	95.38	99.09	97.31	98.27	97.68
DBSMOTE	99.59	98.27	99.09	98.17	98.88	98.8
MWMOTE	98.88	96.24	98.17	97.11	98.16	97.71
OUPS	83.76	79.59	83.4	74.42	82.45	80.72
A-SUWO	99.59	97.67	99.19	98.98	99.18	98.92
NRAS	98.63	98.12	96.85	97.61	97.14	97.67
Developed	97.77	94.57	97.21	92.99	94.13	95.33

However, according to the RON, for normal samples, the Baseline method combined with not only SMC but also KNN exceeds all data augmentation methods. The reason is that classifiers pay more attention to normal samples in the training process, which leads to the problem that fault samples are easily misclassified as normal samples. It is also notable that the safety of gas turbines is extremely important. In other words, to reduce the missed diagnosing rate of fault samples, some misjudgment of normal samples can be bearable, but not vice versa.

4.3. Investigation on performance with different dimensions of latent feature representation

The dimension of latent feature representation is an essential hyperparameter in the developed method. For simplicity, the following analysis takes SMC as an example. Table XII – Table XV show the balanced accuracy, ROE, ROT, and RON of the developed method when using different dimensions. It can be seen that when the dimension of latent feature representation is set to 320, the developed method achieves the best average performance. On one hand, it is worth noting that low dimension (e.g., 64) leads to relatively low performance when K=2, 3, and 5, compared to other dimensions. The reason is that too much important information is lost in the mapping process when using a low dimension, especially for normal samples. On the other hand, the dimensions should not be too high. A high dimension will increase the number of weights and make the training process more difficult, which can negatively impact the diagnosis results of fault samples.

TABLE XII. Balanced accuracy of the developed method under different dimensions

Dimension	Balanced Accuracy (%)					
	K=1	K=2	K=3	K=4	K=5	Average
64	89.34	80.25	92.11	92.78	77.96	86.49
128	91.07	87.47	**98.22**	95.99	79.13	90.38
192	90.21	88.43	92.09	95.75	79.27	89.15
256	90.81	88.65	93.79	92.11	79.61	88.99
320	91.54	88.89	93.57	**96.85**	81.86	**90.54**
384	91.73	88.96	91.14	93.14	80.07	89.01
448	91.64	**89.04**	93.9	89.03	**87.19**	90.16
512	**91.76**	85.28	92.37	89.51	86.6	89.11

TABLE XIII. ROE of the developed method under different dimensions

Dimension	ROE (%)					
	K=1	K=2	K=3	K=4	K=5	Average
64	**100**	**75**	87.5	**100**	**88.57**	90.21
128	**100**	**75**	**100**	**100**	85.71	**92.14**
192	**100**	**75**	81.25	**100**	85.71	88.39
256	**100**	**75**	87.5	85.71	85.71	86.79
320	**100**	**75**	86.25	**100**	85.71	89.39
384	**100**	**75**	78.75	87.14	85.71	85.32
448	**100**	**75**	87.5	74.29	85.71	84.5
512	**100**	62.5	82.5	72.86	85.71	80.71

TABLE XIV. ROT of the developed method under different dimensions

Dimension	ROT (%)					
	K=1	K=2	K=3	K=4	K=5	Average
64	**80**	80	**100**	**100**	60	84
128	**80**	**100**	**100**	**100**	60	88
192	78	**100**	**100**	**100**	60	87.6
256	**80**	**100**	**100**	**100**	60	88
320	**80**	**100**	**100**	**100**	66	89.2
384	**80**	**100**	**100**	**100**	60	88
448	**80**	**100**	**100**	**100**	**80**	**92**
512	**80**	**100**	**100**	**100**	78	91.6

TABLE XV. RON of the developed method under different dimensions

Dimension	RON (%)					
	K=1	K=2	K=3	K=4	K=5	Average
64	88.02	85.74	88.83	78.32	85.31	85.24
128	93.2	87.41	94.67	87.97	91.68	90.99
192	92.64	90.3	**95.03**	87.26	92.09	91.46
256	92.44	90.96	93.86	90.61	93.11	92.2
320	94.62	91.68	94.47	90.56	93.88	93.04
384	95.18	91.88	94.67	92.28	94.49	93.7
448	94.92	92.13	94.21	92.79	95.87	93.99
512	**95.28**	**93.35**	94.62	**95.69**	**96.07**	**95**

4.4. Investigation on the intra-class aggregation and inter-class separation

This paper uses t-distributed stochastic neighbor embedding to provide a perspective on the distributions of original data and latent feature representation, and depicts them in Fig. 5. First, it is easy to see that fault samples highly overlap with normal samples in the original data space, while fault samples are more separable from normal samples in the learned feature space. Second, samples belonging to the same class are dispersed in the original data space, while samples belonging to the same class are gathered better in the learned feature space. For example, EIF samples are gathered around the centroid of their class samples, and are far away from the boundary of normal samples, which can reduce the risk of being misdiagnosed as normal samples.

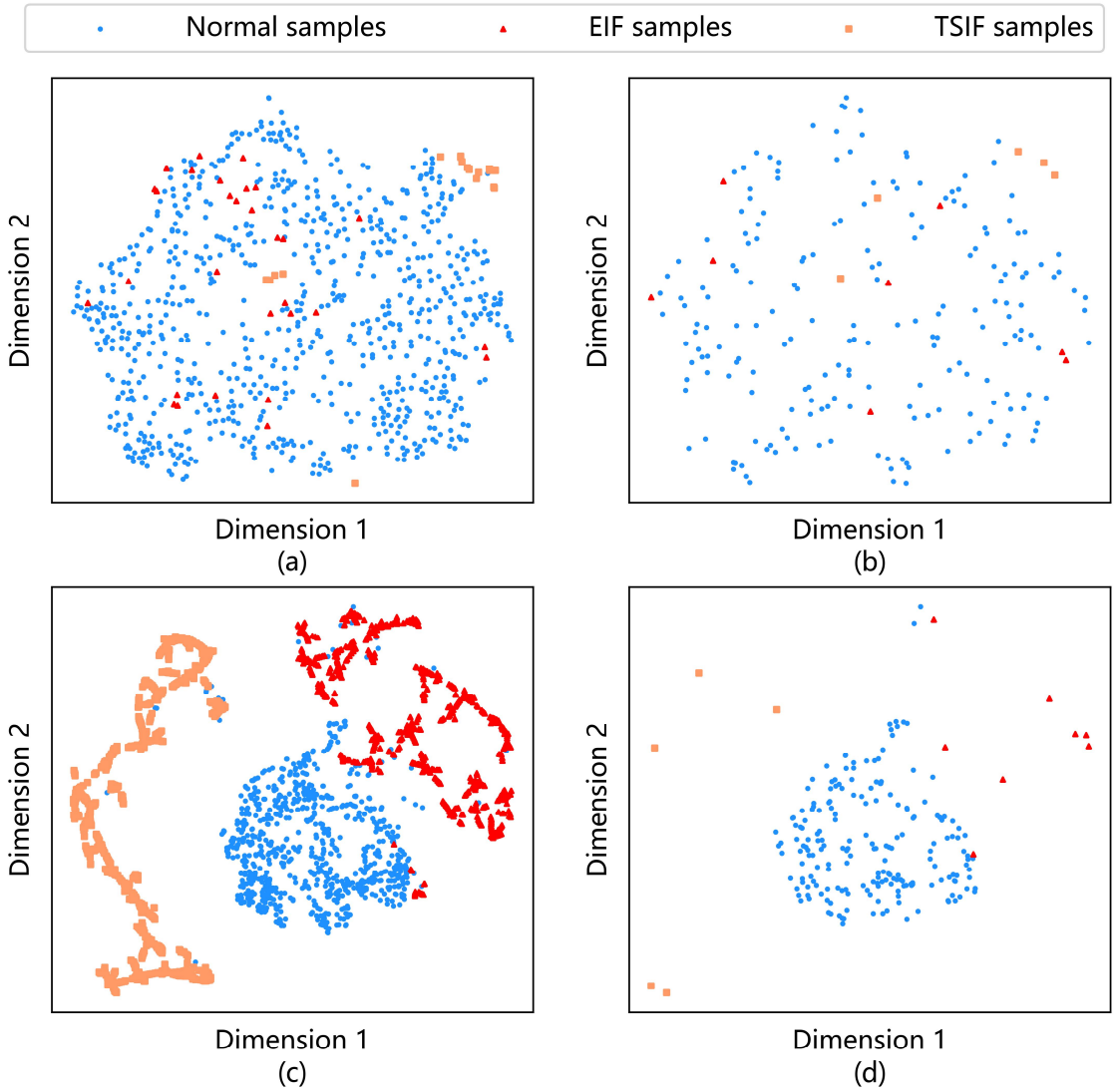

Fig. 5 2D sketch maps of (a) training samples in the original data space; (b) test samples in the original data space; (c) training samples after data augmentation in the learned feature space; (d) test samples in the learned feature space.

5. Conclusion

To address the imbalanced issue in the case of inter-class overlap, this paper develops a novel feature-level data augmentation method that augments fault samples in a learnable feature space. First, the DSMHSA is selected to learn a feature space with large inter-class separation distance, in which fault samples and normal samples are far away from one another. Second, the popular SMOTE is used to generate fault samples in the learned feature space, and then synthetic fault samples and original training samples make up the balanced training set. Finally, the classification model is trained on the balanced data and then diagnoses the health status of gas turbines.

In the experiment, compared with SMOTE (87.01%), SL-SMOTE (84.66%), DBSMOTE (85.97%), MWMOTE (86.53%), OUPS (80.71%), A-SUWO (76.44%), and NRAS (77.99), the performance of the developed method combined with SMC is increased to 90.38% in terms of average balanced accuracy. The scenario of the KNN is also similar. One reason is that the DSMHSA can map the original data into a feature space with large inter-class separation distance, thereby reducing the degree of inter-class overlap. The other reason is that feature-level data augmentation is a better data augmentation strategy.

Finally, imbalanced fault diagnosis of gas turbines is a fast-evolving research topic. Our future efforts will focus on the discovery of new frameworks for learning better feature spaces and performing feature-level data augmentation to further improve the accuracy of gas turbine fault diagnosis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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